

Alien Rescue Game Analytics

159 players | 85,194 log events | 1-hour gameplay session

Player segmentation, early warning system, tool analysis

Data: Liu & Liu (2019), University of Texas at Austin

About This Project

Alien Rescue is a serious game where players must find suitable planets for 6 displaced alien species. Players have access to 10 in-game tools and one hour to solve the problem. This project analyzes the log data of 159 players to answer a simple question from a game studio perspective: why do some players succeed and others fail -- and what can we change in the game to improve that?

Data Sources

File	Rows	What it contains
Log_Raw.csv	85,194	Every in-game action: clicks, notes, navigation
Consoles.csv	5,916	Tool open and close events
Gates.csv	4,460	Zone door crossings
Duration_Characteristics.csv	159	Per-player summaries and survey scores

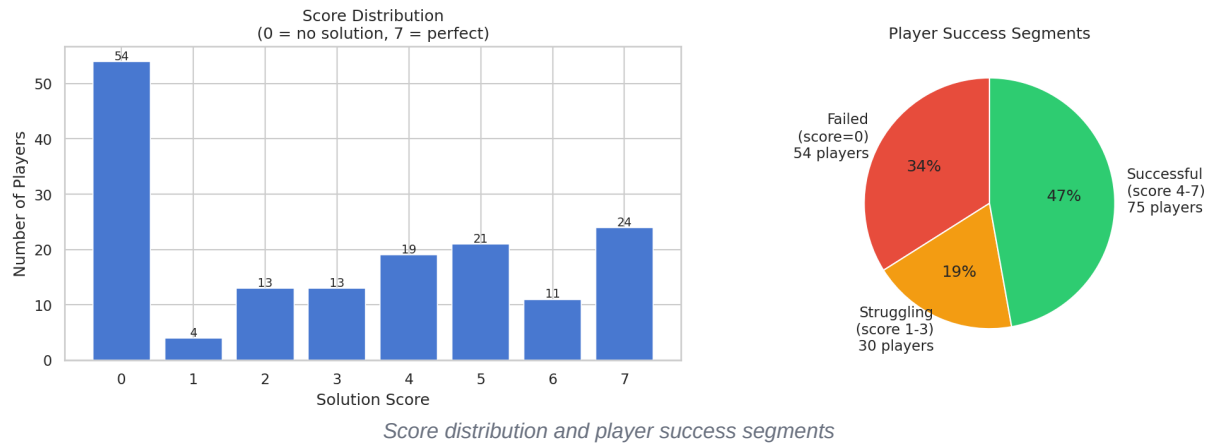
What I Did

1. Loaded and cleaned 4 separate files with inconsistent formats and column names.
2. Built a player profile table by aggregating 85,194 events into one row per player.
3. Explored the data visually to find patterns between behavior and performance.
4. Applied K-Means clustering to identify 3 distinct player types.
5. Tested whether the first 20 minutes of play can predict final score.
6. Evaluated all 10 in-game tools for adoption rate and value contribution.

Performance Overview

The first thing I wanted to understand was: are players actually succeeding? Solution score ranges from 0 (no alien saved) to 7 (all 6 saved). The result was surprising.

Player Performance Overview

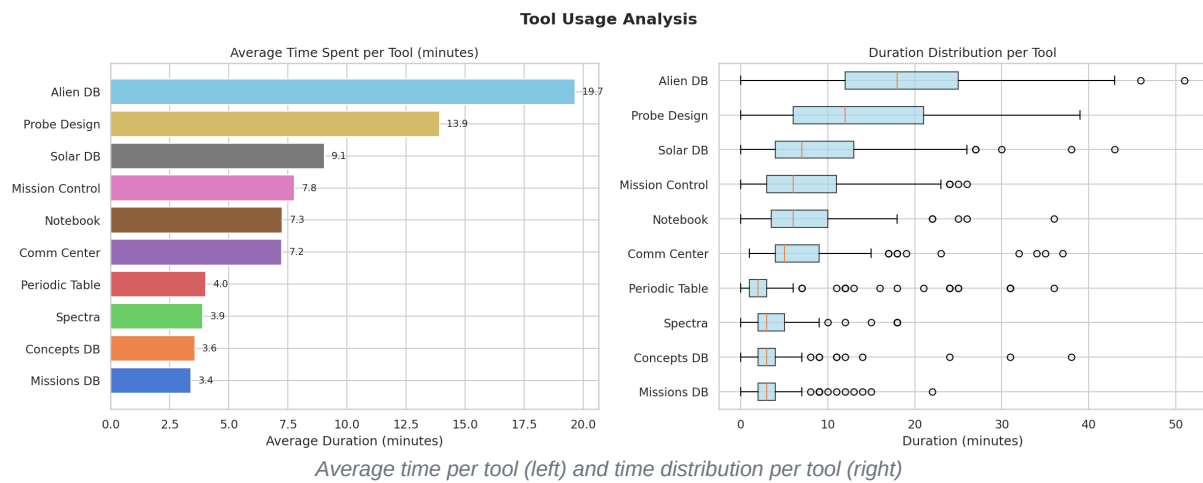


What this tells us

33% of players scored exactly 0 after a full hour of gameplay. They were not rushing or skipping -- they played the entire session and still could not save a single alien. This is the clearest signal in the entire dataset: something in the game is preventing a third of players from even getting started. This is likely an onboarding or discoverability problem.

Tool Usage

Players have access to 10 tools. Before looking at which tools are helpful, I wanted to see where players actually spend their time.

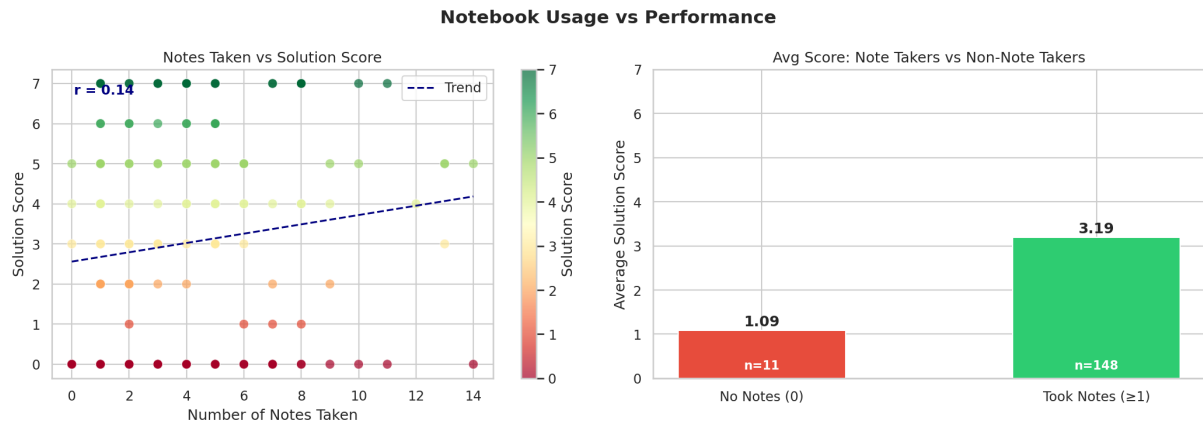


What this tells us

Alien DB is the most used tool by a large margin -- players spend an average of almost 20 minutes there. Probe Design, which is the core problem-solving mechanic of the game, receives only about 14 minutes on average. The box plots show that time spent in most tools varies widely across players, meaning some players discover and use certain tools while others ignore them entirely.

Does Note-Taking Help?

The game includes a Notebook tool where players can take and organize notes. I wanted to test whether players who use it actually perform better.



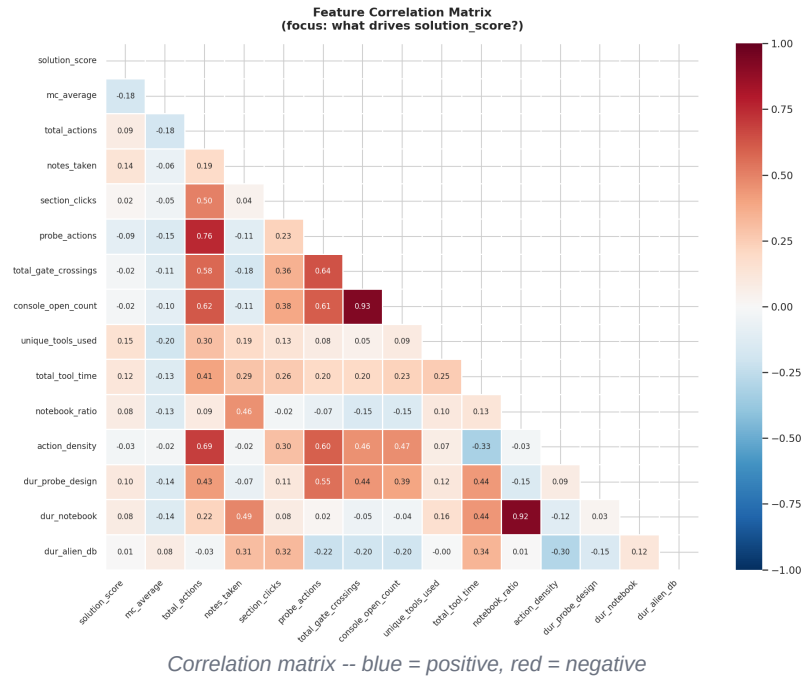
Notes taken vs solution score, and average score by note-taking group

What this tells us

Players who take at least one note score noticeably higher on average. The correlation is positive ($r = 0.14$) but not very strong, which means note-taking alone is not enough -- but players who organize their findings tend to do better. The problem is that most players take very few notes or none. The Notebook feature exists in the game but is not being used effectively.

Feature Correlations

Before building the segmentation model, I looked at which player behaviors and characteristics correlate most with solution score.



Strongest predictors of solution score

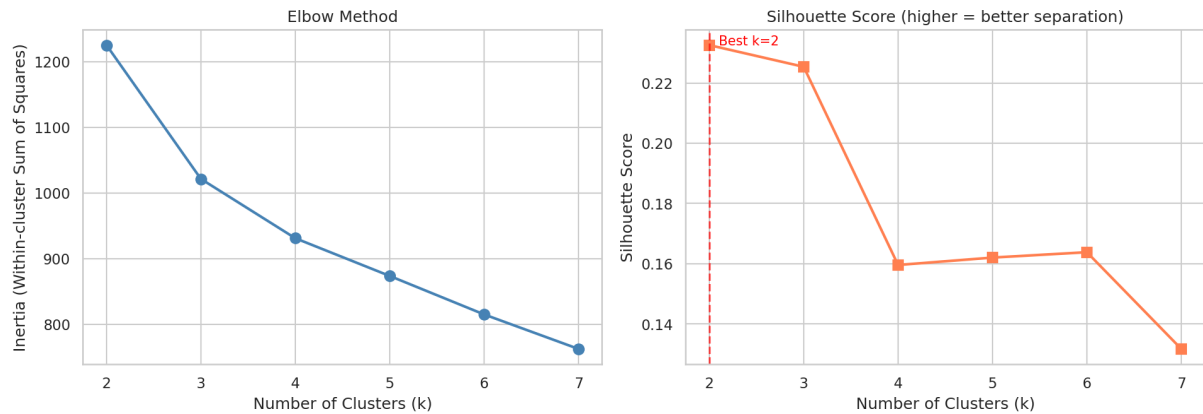
Feature	Correlation	Interpretation
unique_tools_used	+0.154	Players who explore more tools score higher
notes_taken	+0.140	Note-taking is associated with better performance
total_tool_time	+0.119	More engaged players tend to do better
dur_probe_design	+0.104	Time in Probe Design pays off
mc_average	-0.177	Counterintuitive -- explored in segmentation

The negative correlation between metacognition score and solution score was unexpected. This becomes clearer when we look at the player segments -- it is not that metacognition hurts performance, but that high-metacognition players are distributed unevenly across segments.

Player Segmentation

I used K-Means clustering on 10 behavioral features to group players. The goal was to find out if there are meaningfully different types of players in the data -- and if so, what makes them different.

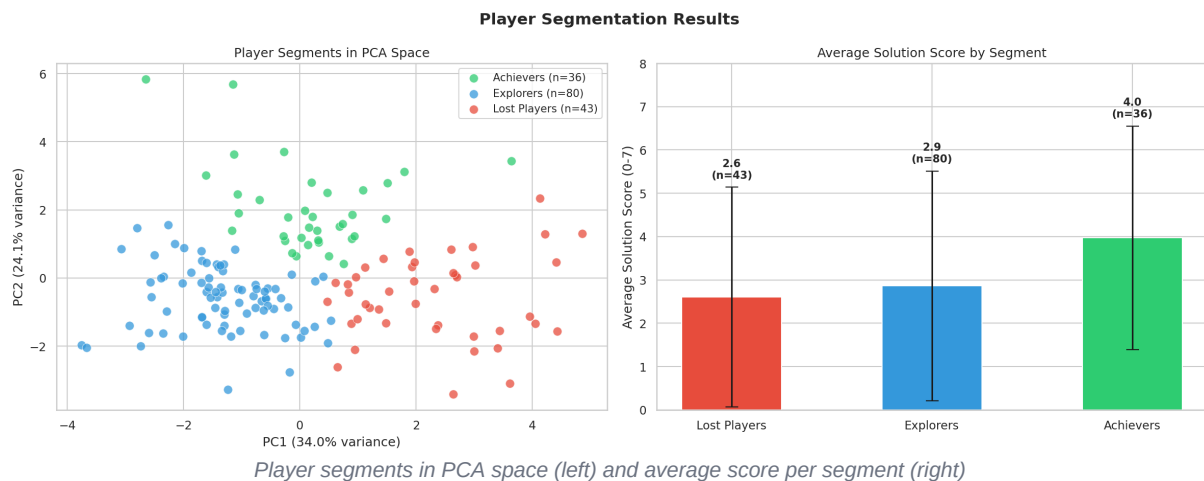
Finding Optimal Number of Segments (k)



Elbow method and silhouette score to find the optimal number of clusters

Both methods pointed toward k=2 or k=3. I chose k=3 because three segments make more practical sense for a game studio than two.

The 3 Player Types



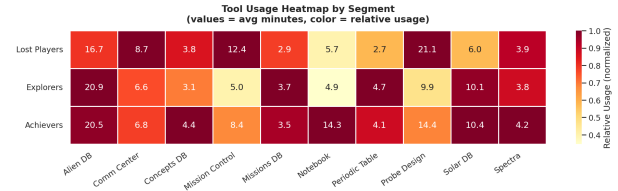
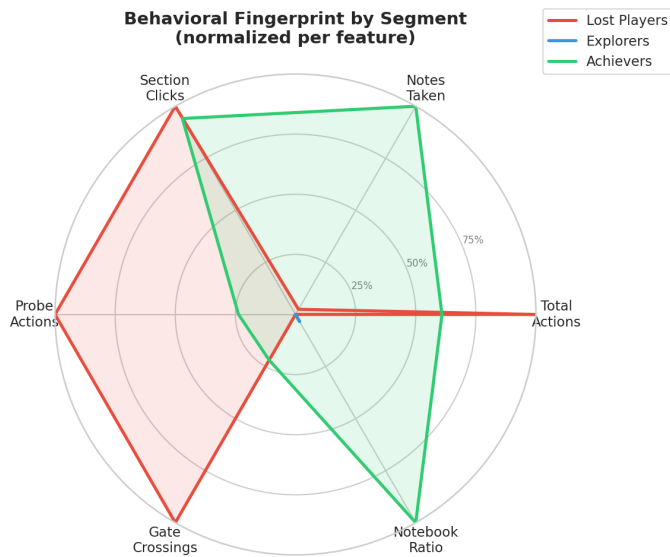
Segment	Count	Avg Score	Key Behavior
Achievers	36	3.97	Take 2x more notes, spend 16% of time in Notebook
Explorers	80	2.86	Largest group, lowest activity, passive engagement
Lost Players	43	2.60	Most actions (748 avg) but scattered, no clear strategy

The most important finding

Lost Players generate the most clicks of any group -- 748 actions on average -- but they score the lowest. They are not passive or disengaged. They are clicking constantly but without direction. This tells us the game is not giving players enough structure to guide their exploration. Activity alone does not lead to success.

Behavioral Fingerprints

The radar chart below shows the normalized behavioral profile of each segment. It makes it easy to see what Achievers do differently from the other two groups.

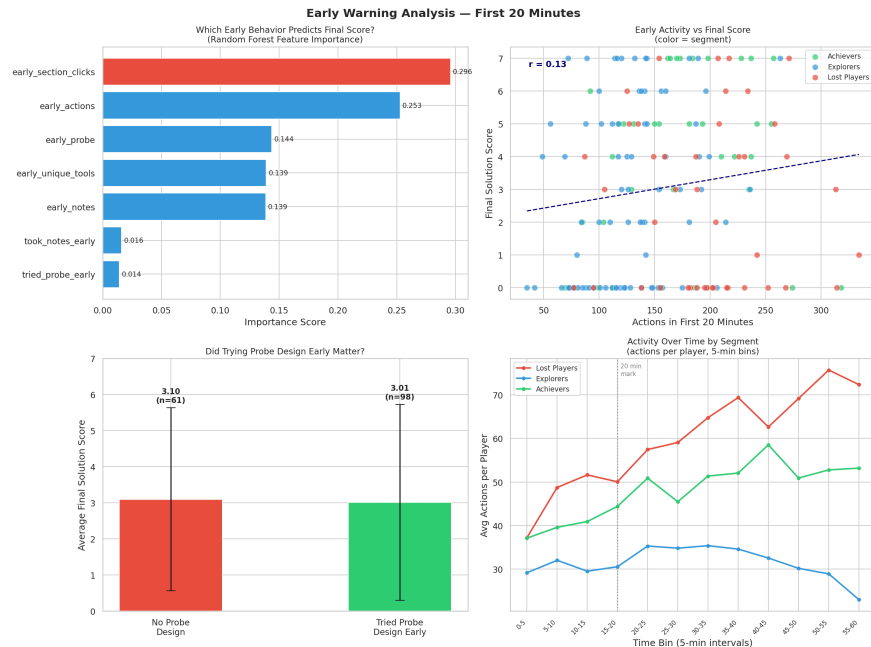


Radar: behavioral fingerprint per segment -- Heatmap: tool usage (minutes) per segment

Achievers stand out clearly on the notes axis. The heatmap confirms that Lost Players spend more time in Probe Design than Explorers but still score lower -- suggesting they enter the tool but do not know what to do with it.

Early Warning System

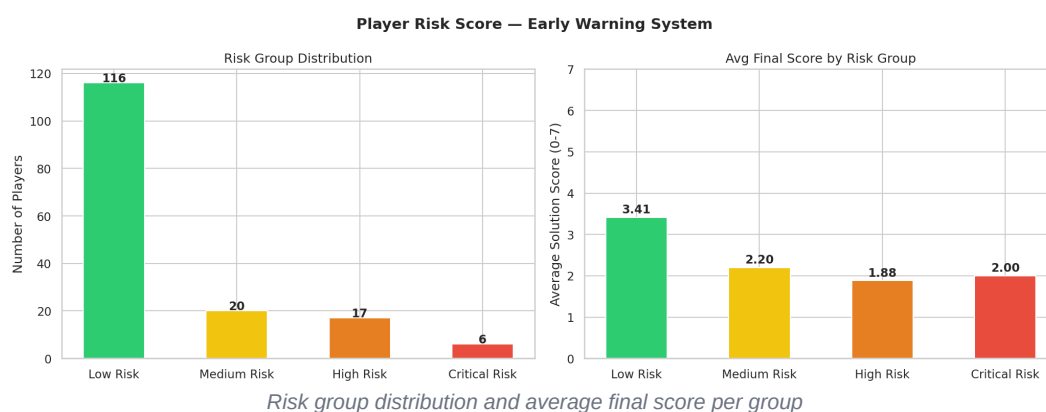
Can we tell which players are going to struggle -- before they fail? I isolated the first 20 minutes of each session and extracted 7 behavioral signals. I then tested whether those signals could predict final score using Random Forest.



Top left: feature importance -- Top right: early actions vs final score by segment -- Bottom: activity over time by segment

Result: R-squared = -0.07

The model could not reliably predict final score from the first 20 minutes alone. This is an important negative finding. It means the critical moment in this game is not the opening -- it is somewhere in the middle of the session. Looking at the activity-over-time chart, Explorers show a clear drop in activity after the 20-minute mark. This is where they disengage.

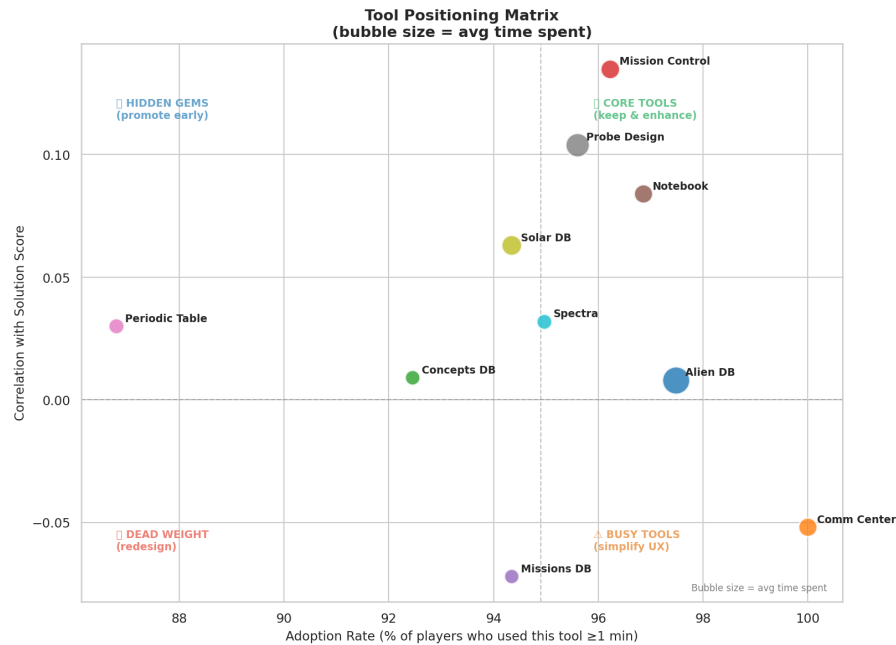


Risk group distribution and average final score per group

Even though the model failed, a simple 4-factor risk score (low early activity + no notes + no Probe Design + few tools) still separates players meaningfully. Low Risk players average 3.41 while High Risk players average 1.88. This could be implemented in a real game to trigger a hint at the 20-minute mark.

Tool Analysis

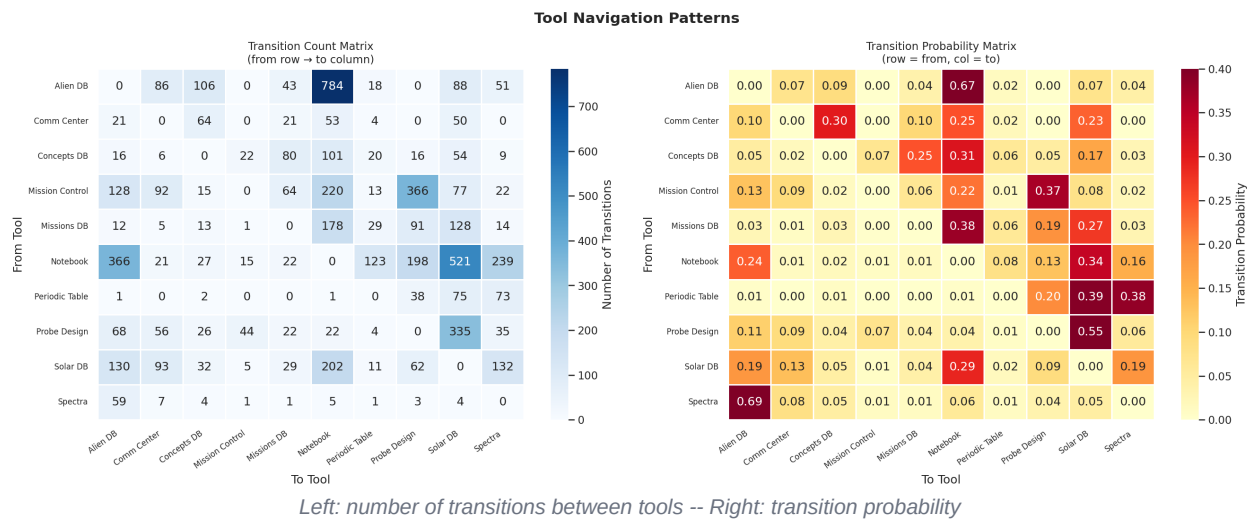
I evaluated each tool on two dimensions: adoption rate (what percentage of players used it for at least 1 minute) and correlation with solution score (does using it actually help?). Bubble size represents average time spent in the tool.



Category	Tools	What it means
Core Tool	Mission Control, Probe Design, Notebook	High adoption and positive correlation. Keep and surface early.
Hidden Gem	Solar DB, Periodic Table, Concepts DB	Positive correlation but lower adoption. Add to onboarding.
Busy Tool	Comm Center	100% adoption but negative correlation. Players spend time here without benefit.
Dead Weight	Missions DB	High adoption but negative correlation. Needs content review.

How Players Move Between Tools

I also looked at tool transitions -- when a player leaves one tool, which tool do they go to next? This reveals navigation patterns.



Most traffic flows between Alien DB, Solar DB, and Mission Control. Players are looping within this triangle. Probe Design receives fewer incoming transitions than expected for the most critical tool in the game. Players are not finding their way to it naturally.

Conclusions

Here is a summary of the main findings and what I would recommend if this were a real game studio project.

Finding	Recommendation
33% of players score 0 after a full session	Redesign onboarding. Players are not discovering the tools they need.
Note-taking predicts success but most players do not use Notebook	Introduce Notebook in a tutorial within the first 10 minutes.
Lost Players click the most but have no strategy	Add a hint system that triggers after many actions without progress.
First 20 minutes do not predict final score	Focus intervention at the mid-game point, not the beginning.
Probe Design is underused despite strong positive correlation	Make Probe Design more visible. Surface it in the first 5 minutes.
Missions DB has a negative correlation with score	Audit the content. Players may be spending time there unproductively.

This project was built on a publicly available dataset. All findings are exploratory. Recommendations should be validated through A/B testing before being implemented in a live game.